TrustCharge: Reliability-Aware Long-Trip Charging Planning and Port Reservation for Battery Electric Vehicles Using Live Open Data

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***Abstract*—Long-distance travel with battery electric vehicles (BEVs) is limited by range anxiety, long charging times and, increasingly, queues at fast chargers. Planners that model waiting time typically rely on survey-derived or simulated occupancy, and they implicitly trust that a listed station actually works. This paper presents TrustCharge, a web platform for Dutch BEV drivers that turns live open data (NDW charge-point feed and tariffs, TomTom availability, OpenChargeMap status, RDW vehicle registry, Smartcar telemetry) into a per-bay view of every station, scores station reliability from observed evidence (withholding the score when no evidence exists), and plans long trips with a dynamic program in the spirit of Bhat et al. that minimizes driving, waiting, charging and expected fault delay. A ticket-style reservation subsystem with tiered membership turns waiting from a forecast into a schedulable resource. A generative assistant is confined to explanation; every number shown to the driver comes from deterministic code. In a synthetic 500 km corridor study (300 corridors, 30 replications each), the proposed planner shortened mean door-to-door time by 25.0 min (7.0%) against an on-need greedy policy and by 6.5 min against a dynamic program ignoring waiting and reliability, and reduced queue time by 40% and fault delay by 36% relative to the two nearest baselines. Results are simulation-based; field validation is future work, and we state which modules are implemented and which are still in integration.**

***Index Terms*—Battery electric vehicles, charging planning, waiting time, dynamic programming, station reliability, port reservation, open data**

I. Introduction

Battery electric vehicles are displacing combustion vehicles quickly, and the Dutch public charging network is among the densest in Europe. Yet a long trip remains uneven: a driver must decide where to stop, how much energy to take on, and whether a listed charger will be free and working on arrival. Fast-charging stops are a large share of trip time, and when stations are crowded the queue can rival the charging itself. Selecting stations with waiting time in mind directly affects total trip time [1], and knowing the expected wait can justify a detour to a quieter station [2].

Two obstacles stand between waiting-aware planning research and a driver's phone. First, waiting-time inputs in the literature are typically parameterized from surveys or simulation, not from the live state of the network. Second, availability feeds say whether a connector is listed as free, not whether the station has been dependable over recent weeks; a driver routed to a faulty charger pays in minutes, not in a lower star rating. Consumer apps show live maps but do not plan multi-stop trips against forecast queues or let a driver secure a bay, and general-purpose language models can produce fluent but unverifiable claims about prices, availability or reliability.

We therefore propose TrustCharge, built on three ideas. The data layer converts the NDW live feed and partner sources into a per-bay picture of each station and records observations from which a reliability score is derived, left empty when there is no evidence. The planning layer adapts the multi-stage formulation of Bhat et al. [1] to a state-of-charge grid and adds an expected fault-delay term. The reservation layer lets drivers book a specific bay, with fees, horizon and cancellation terms set by a membership tier. A generative assistant sits outside the trust boundary and may only phrase results already produced by deterministic code.

II. Related Works

Charging-aware planning. Baum et al. engineer constrained shortest-path algorithms for BEVs with contraction hierarchies and A* [3]. Most approaches assume a full recharge at each stop, and few consider partial charging [2]. Because lithium-ion charging slows markedly beyond roughly 80% state of charge [4], a realistic planner must decide how much to charge, not only where. Deep reinforcement learning planners handle stochastic conditions.

Grid-side and AI-based smart charging This covers the first six studies from your PPT table. It describes their methods (station allocation on the IEEE 69-bus network, Honey Badger Algorithm, ISO 15118, ML with V2G, deep-learning load balancing with blockchain, GRU with Harris Hawk optimization). It then explains what they leave out: booking, trust evaluation, prepaid access, and time-dependent failures.

Reservation and data-driven studies This covers the reservation studies, the business-intelligence model, federated occupancy prediction and the cross-lingual review-based reliability work. It ends by contrasting them with TrustCharge,

which derives reliability from observed port evidence and ties it to bookable bays

Waiting-time forecasting and coordination. Bhat et al. formulate long-trip planning as a mixed-integer dynamic program minimizing total trip time, with a waiting-time forecast parameterized from open US household travel survey data [1]. Schoenberg and Dressler estimate waiting from current utilization, announced stops and statistical utilization per hour [2]; De Weerdt et al. combine announced intentions with historical data [5]; reservation-based planners let vehicles book a slot but need provider cooperation [6]. In all of these, waiting is an input to be forecast; missing is a source built from the live state of a real national network and a mechanism that lets a driver act on it.

Infrastructure, reservation and data-driven studies. Table I summarizes twelve further studies reviewed for this project. Station-allocation and smart-grid work [7]–[12] optimizes where and when to charge from the grid side, but offers no booking, trust evaluation or per-station reliability. Reservation studies [13], [14] let a driver book before arrival but forecast neither waiting nor demand. Data-driven studies [15]–[17] predict schedules, occupancy or reliability from reviews, yet depend on large real-time datasets and say little about real-time operation.

Taken together, the studies leave four gaps: no single system combines evidence-based reliability with cross-network reservation; reservation systems do not forecast waiting; optimization and AI approaches rarely support real-time slot booking or prepaid access; and real-world, large-scale validation is largely missing. Retrieval-grounded assistants reduce, but do not remove, unsupported numerical claims [18], so we let the language model explain but not compute. TrustCharge is not new optimization theory: it adopts the formulation of [1], borrows the utilization-forecast idea of [2], and contributes a deployable pipeline in which forecast, reliability evidence, bay-level reservation and interface come from one live data layer.

III. Problem Definition

Existing charging apps such as Tata Power EZ Charge, Statiq, ChargeZone and Bolt.Earth locate stations, navigate to them and take payments, but each works independently and serves only its own network. A driver is left with separate applications and wallets, no way to compare stations across networks, no evidence of whether a charger works, and little help with queues. TrustCharge serves EV drivers and fleet operators, charging-network operators and transport authorities, starting with NDW open data in the Netherlands. At the first step of this work, we focus on two questions:

- What evidence can be turned into a trustworthy reliability signal for a listed station, and what should be shown when there is none?
- How can a long trip and a bay booking be planned against forecast waiting time and expected fault delay?

Formally, let a corridor have candidate stations $1, \ldots, N$, each with connector groups, live state, a reliability R_i (or none) and a waiting forecast $\hat{w}_i(t)$. For a vehicle with usable energy E_b, reserve $\sigma_{\min}$ and departure time t_0, we seek the stops (i, σ, σ') that minimize expected door-to-door time. A useful platform must (i) build a per-bay view from live open data, (ii) derive reliability only from observed evidence, (iii) let the driver reserve a bay under server-enforced terms, and (iv) keep every number traceable to deterministic code.

IV. TrustCharge Framework

Fig. 1 presents the overview of the framework.

A. System Architecture

A React 19 single-page client talks to a Node.js Express 5 API that authenticates requests with Firebase ID tokens, applies per-minute rate limits and persists state in embedded SQLite. The API pulls live data from nine external services: the NDW DOT-NL feed and OCPI tariffs, TomTom availability, OpenChargeMap status, the RDW registry, Smartcar telemetry, OSRM routing, Nominatim geocoding and Mollie payments (plus Firebase authentication and the Gemini assistant). Deterministic services convert these inputs into stations, ports, reliability scores, plans, reservations and wallet balances. The Gemini assistant is invoked last and receives only already-computed context.

B. Data Ingestion and the Port Model

The NDW feed reports, per station, availability groups: for a connector type and power level, the number of free and total connectors, but no stable identifier per physical socket. TrustCharge therefore models a station as connector groups g with n_g bays numbered $1, \ldots, n_g$, marking the first a_g bays free when live state applies. A station is OUT OF SERVICE only when every known port is faulty, AVAILABLE when at least one port is free, OCCUPIED otherwise, and UNKNOWN if no counts exist. Prices are shown only when an OCPI tariff was actually matched. OpenChargeMap and TomTom entries are matched to NDW stations within 200 m and 100 m respectively, each contributing its own observation record. Because a bay number is a slot rather than a hardware identifier, a reservation guarantees exclusivity inside TrustCharge, and the interface says so.

C. Calculation of Reliability

When a station is normalized, the system may record an observation (status, free/total/faulty ports, port-health percentage, source), at most one per station and source every five minutes so that counts depend on time, not page loads. Let V be the valid observations over the last 30 days (at most 1000), with mean port health $\bar{h} \in [0, 100]$, N_{op} observations AVAILABLE or OCCUPIED, and N_f OUT OF SERVICE. Define

$$\rho_h = \bar{h}/100, \qquad \rho_o = \frac{N_{op}}{N_{op} + N_f}, \tag{1}$$

$$E = 0.75\,\rho_h + 0.25\,\rho_o, \tag{2}$$

$$C = 0.75\,E + 0.25\,\bar{r}/5, \tag{3}$$

$$R = 100 \cdot \max\bigl(0,\ C - \min(0.15,\ 0.01\,n_{neg})\bigr), \tag{4}$$

TABLE I
COMPARISON OF RELATED STUDIES

Ref.	Study (year)	Methodology	Gap identified
[7]	Multi-objective station allocation (2023)	Multi-objective optimization of station and renewable placement (IEEE 69-bus)	No real-time booking, trust evaluation or predictive availability
[8]	EV integration into smart grid (2024)	Improved Honey Badger Algorithm for V2G and G2V energy management	No slot reservation, real-time availability or prepaid booking
[9]	Smart charging at public infrastructure (2024)	ISO 15118 communication architecture at a public charging hub	No reservation, route planning or reliability prediction
[10]	AI-based smart charging, Malaysia (2024)	ML, predictive analytics and demand response with V2G optimization	Limited deployment and local datasets; no large-scale AI
[11]	AI-driven load balancing and grid integration (2025)	Deep-learning demand forecasting, dynamic load balancing, blockchain V2G	Centralized systems are vulnerable; limited scalability
[12]	AI-driven reliability-based planning with renewables (2025)	GRU load forecasting, Harris Hawk optimization, Bus Reliability Index	Ignores time-dependent failures and charging uncertainty
[13]	Reservation-based request distribution (2023)	Book a station before arrival; tested on real data	Advance booking needed; no AI waiting-time or demand prediction
[14]	Reserving decision model and route plan (2018)	K-shortest-path station choice by traffic, distance, cost, availability	No AI/ML; only route planning and reservation
[15]	EV charging: business intelligence model (2025)	AI/ML on EV and station data to recommend charging schedules	Needs large real-time data; hard to scale; secure sharing required
[1]	Long-trip planning with waiting forecast (2025)	Mixed-integer DP with Monte Carlo waiting-time forecast	Fixed traffic assumptions; no real-time traffic or dynamic pricing
[16]	Federated learning for station occupancy (2025)	Federated learning evaluated by MAE, RMSE, MAPE; preserves privacy	Needs testing on larger, diverse real-time networks
[17]	Cross-lingual reliability of charging infrastructure (2023)	BERT and machine translation on multilingual station reviews	Depends on translated reviews; weak real-time support
This work	Live per-bay data, reliability, planner, reservation	Evidence-based reliability score, waiting- and fault-aware DP, transactional bay booking	Synthetic evaluation only; learned prediction not yet implemented

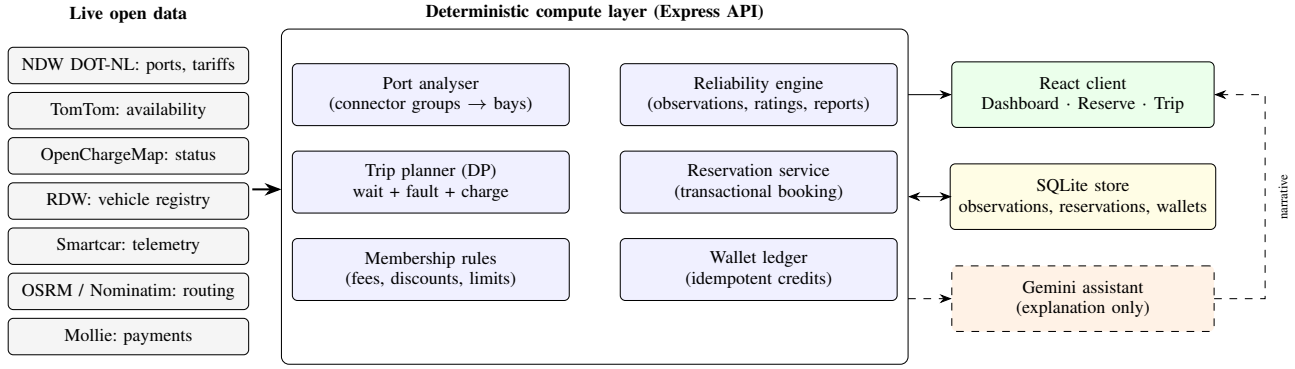


Fig. 1. TrustCharge architecture. Live sources feed a deterministic compute layer that owns every number shown to the driver; the dashed generative component receives only already-computed context and returns narrative text.

where $\bar{r}$ is the mean customer rating (1–5) and n_{neg} the number of negative reports; if a term is missing the remaining terms are used unchanged. The star value is $R/20$. A confidence $\kappa = \min\{100, \min\{100, 2|V|\} + \min\{20, 2n_r\}\}$ grows with observations and ratings n_r, and a label (early, observed, established, strong) is derived from $|V|$ and κ. When $|V| = 0$ no score is produced and the interface shows that data are still being collected.

D. Calculation of Waiting Time

For station i with bays B_i and planned arrival t, the forecast waiting time is the time until the earliest bay becomes free:

$$\hat{w}_i(t) = \max\Big(0,\ \min_{b \in B_i} f_b(t) - t\Big) + \hat{w}_i^{stat}(t), \qquad (5)$$

where $f_b(t)$ is t if bay b is free at t and otherwise the end of the reservation (or live occupancy) covering t, following the availability logic for announced stops in [2]. Live occupancy is applied only within a 30-minute window; beyond it only reservations block a bay. The statistical term $\hat{w}_i^{stat}(t)$ represents unannounced demand and is obtained from the historical hourly utilization $u_i(h)$ by the phantom-stop device of [2], inserting short synthetic stops with period $T = t_{chr}/(u\,n)$ for n charge points. In the current build the live and reservation components are available; the statistical component is scheduled for integration (Section V).

E. Waiting- and Reliability-Aware Trip Planning

The origin is node 0, candidate stations are nodes $1, \ldots, N$, and the destination is node $N{+}1$, with positions x_i and

Algorithm 1 Waiting- and Reliability-Aware Charging Plan

Require: Stations $1..N$ with $x_i, P_i, r_i, \hat{w}_i(\cdot)$; vehicle $(E_b, c, P_v, \sigma_0, \sigma_{\min})$; departure t_0
$\tau_0(\sigma_0) \leftarrow 0$; all other $\tau_i(\sigma) \leftarrow \infty$
for $i = 0$ to N **do**
  for each $\sigma' \in \{0..100\}$ **do**
    $A(\sigma') \leftarrow \min\{\tau_i(\sigma'), \min_{\sigma<\sigma'}[\tau_i(\sigma) + K_i(\sigma, \sigma', t_0 + \tau_i(\sigma))]\}$ ▷ $i = 0$: no charging
  end for
  for $j = i + 1$ to $N + 1$ **do**
    for each σ' with $\sigma' - \delta_{ij} \geq \sigma_{\min}$ **do**
      relax $\tau_j(\sigma' - \delta_{ij}) \leftarrow \min\{\cdot, A(\sigma') + d_{ij}/v\}$; store back-pointer
    end for
  end for
end for
$\sigma^\star \leftarrow \arg\min_\sigma \tau_{N+1}(\sigma)$; **return** stops (i, σ, σ')

distances $d_{ij} = x_j - x_i$. The state is (i, σ) with $\sigma \in \{0, \ldots, 100\}$ the arrival state of charge in percent. Following [1], the decision at a station combines a binary stop choice and a charge target $\sigma' \geq \sigma$; discretizing σ at 1% makes the enumeration exact. Let $\epsilon = E_b/100$ be the energy per percent, P_i the station power, P_v the vehicle limit, and $g(m) = 1$ for $m < 80$, $g(m) = 1 - 0.8(m - 80)/20$ for $m \geq 80$ the charge-curve taper [4]. The cumulative charge time is

$$\Theta_i(\sigma) = \sum_{m=0}^{\sigma-1} \frac{\epsilon}{\min(P_i, P_v)\, g(m)}. \tag{6}$$

The cost of stopping at station i when arriving at clock time t with charge σ and leaving with $\sigma' > \sigma$ is

$$K_i(\sigma, \sigma', t) = t_p + \hat{w}_i(t) + (1 - r_i)T_f + \Theta_i(\sigma') - \Theta_i(\sigma), \tag{7}$$

with plug-in overhead t_p, reliability $r_i = R_i/100$ and fault penalty T_f (time lost when a station is unusable). Let $\delta_{ij} = \lceil 100\, c\, d_{ij}/E_b \rceil$ be the charge percentage needed to drive from i to j at consumption c (kWh/km), $\sigma_{\min}$ the reserve, and $\tau_i(\sigma)$ the smallest elapsed time at which (i, σ) can be reached. Then

$$\tau_j(\sigma' - \delta_{ij}) = \min_{\sigma \leq \sigma'} \Big\{ \tau_i(\sigma) + K_i\big(\sigma, \sigma', t_0 + \tau_i(\sigma)\big) + d_{ij}/v \Big\}, \quad \sigma' - \delta_{ij} \geq \sigma_{\min}, \tag{8}$$

where $K_i = 0$ if $\sigma' = \sigma$ and t_0 is the departure time. The plan is $\min_\sigma \tau_{N+1}(\sigma)$, recovered by back-pointers (Algorithm 1). Keeping only the earliest elapsed time per state is exact when arriving earlier never leads to a later departure (FIFO), which holds when $\hat{w}_i(t)$ changes slowly relative to t. Stations without a score use an internal conservative prior for r_i, never displayed as a rating. The inner minimization costs $O(|\Sigma|^2)$ per node and transitions $O(N|\Sigma|)$ with $|\Sigma| = 101$, so a 14-station corridor plans in milliseconds.

F. Bay-Level Reservation and Membership

The reservation page presents a station as a ticket-booking map. Connector groups are rows (connector type, power, tariff) and each bay is a tile that is FREE NOW, OPEN, IN USE, BOOKED or FAULT. Stations are ranked by reliability first, and an auto-pick control proposes the fastest bookable bay at the most reliable station. The reservation fee is EUR 1.00 per 30 minutes before discount. Table II lists the tiers; benefits are stored server-side and enforced at booking, at refund on cancellation, and at billing, where the energy discount is applied to the recomputed session cost, never to a client-supplied cost.

TABLE II
MEMBERSHIP TIERS AND ENFORCED BENEFITS

	Standard	**Plus**	**Pro**
Price per 30 days	free	EUR 4.99	EUR 9.99
Wallet credit included	–	EUR 10	EUR 25
Reservation fee discount	0%	50%	100%
Energy discount	0%	5%	12%
Book ahead (h)	24	72	168
Active reservations	1	2	4
Longest reservation (min)	60	90	120
Free cancel until (min before)	60	30	10

Booking is a single transaction: the server validates the duration ($\in \{30, 60, 90, 120\}$ min) and booking horizon against the driver's tier, re-reads the live station and verifies that the bay exists and is selectable. Inside an immediate write transaction it then re-checks overlap on the bay, overlap with the driver's own reservations and the active-reservation limit, checks the wallet balance, and only then debits the fee and inserts the reservation; any failed check rolls back. Two drivers therefore cannot hold the same bay for overlapping windows, and a retry cannot debit twice.

G. Generative Boundary

The Gemini assistant receives the verified vehicle, live telemetry, the selected station and a capped list of nearby stations, with rules forbidding invented stations, prices, availability, telemetry, reliability or ratings, and treating the supplied context as data rather than instructions. The AI station recommender may only return the identifier of a candidate the server itself filtered as available and within live estimated range; any other identifier is discarded and a local score (reliability, or a neutral value if unknown, minus a mild distance penalty) is used instead.

V. EXPERIMENTAL EVALUATION

In this section we present the implementation of the proposed system and its evaluation. First we describe the implementation and module validation. Then we work through a synthetic corridor case study.

The frontend uses React with Vite and react-leaflet on OpenStreetMap tiles; the backend uses Node.js, Express and the built-in `node:sqlite` module, with Firebase ID-token verification and in-memory rate limiting. Data come from NDW [19], RDW [20], TomTom [21], OpenChargeMap, Smartcar [22], OSRM [23], Nominatim and Overpass; payments use Mollie hosted checkout [24]; the assistant uses Gemini through its REST interface with a fallback model. A modest server (Core i3, 4 GB RAM) suffices. Engineering lessons were mainly about trust: Mollie webhooks and client polling can

both observe the same payment, so each payment type has its own ledger with a credited flag and credit is granted inside an immediate transaction; renewal extends from the later of now and current expiry so benefits are never briefly withdrawn; checkout return addresses are restricted to the configured frontend; per-station, per-source sampling every five minutes keeps busy stations from looking better documented than quiet ones; and Smartcar tokens are encrypted at rest (AES-256-GCM) with HMAC-signed, expiring OAuth state.

VI. Limitations.

The study is synthetic: stations, demand and reliabilities are generated, the forecast is assumed unbiased and independent between stations, and realized waits are exponential, whereas real queues are correlated and depend on other drivers' choices, including those using the same planner. The reliability score is an observed-availability indicator, not a calibrated failure probability, so mapping $1 - r_i$ to a fault probability is a modelling assumption. A fault costs a fixed 25 minutes and does not strand the vehicle. The corridor is one-dimensional with no detours and constant energy use per kilometre, and bays are slots, not physical socket identifiers. Absolute numbers are not comparable with [1] or [2]; only the ordering of strategies and its mechanisms should be read from Table IV.

VII. Future Work

In future, we would like to extend the work in three stages. Near-term: complete the reservation endpoints and tier enforcement, add automated tests, release unclaimed bays after a grace period, and keep collecting station observations, since 3,990 samples are too thin to train on. Mid-term: integrate the planner with the live station list and OSRM routes, compute the statistical utilization term from stored observations, treat reservations as announced stops [2], re-plan at every stop, and train AI models for charger reliability and queue-time prediction to feed station ranking and the assistant. Longer-term: calibrate the reliability score against observed failures, measure forecast error against real arrivals, move from simulated to live Smartcar vehicles, onboard more providers and regions beyond the Netherlands, and run a field study reporting time saved, forecast accuracy and reservation no-show rates.

VIII. Conclusion

This paper presented TrustCharge, which joins live Dutch charging data, an evidence-based reliability score, a waiting- and reliability-aware dynamic programming planner and a bay-level reservation model with tiered membership. Every quantitative claim stays deterministic and traceable, with the language model confined to explanation. In a synthetic corridor study the planner reduced door-to-door time, and in particular time lost to queues and faults, relative to an on-need policy and to planners ignoring waiting and reliability. The main engineering lessons concerned idempotent payments, honest handling of missing data, and the difference between listed and dependable availability. Field validation and learned reliability and queue prediction remain to be completed.

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